



DWF_dp_mult_comb function

Combined Unsigned/Signed Multiply

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Description

Revision History

The DWF_dp_mult_comb function performs combined (switchable) unsigned/signed multiplication of the two arguments a and b. Argument a (b) is interpreted as signed if argument a_tc (b_tc) is 1, otherwise as unsigned. The result must be interpreted as signed if argument a_tc or b_tc (or both) is 1 (= signed multiplication), otherwise as unsigned (= unsigned multiplication).

Table 1-1 Function Names

Function Name	Description
DWF_dp_mult_comb	VHDL combined multiply (std_logic_vector/unsigned/signed arguments)
DWF_dp_mult_comb	Verilog combined multiply
DWF_dp_mult_comb_tc	Verilog combined multiply (signed arguments)

Table 1-2 Argument Description

Name	Type	Width / Values	Description
a	Vector	<i>a_width</i>	Input multiplier
a_tc	Bit	1	Two's complement control for multiplier 0 = Unsigned 1 = Signed
b	Vector	<i>b_width</i>	Input multiplicand
b_tc	Bit	1	Two's complement control for multiplicand 0 = Unsigned 1 = Signed
DWF_dp_mult_comb	Vector	<i>a_width+b_width</i>	Returned value

Table 1-3 Parameter Description (Verilog)

Parameter	Values	Description
a_width	≥ 1	Word length of input a
b_width	≥ 1	Word length of input b

Verilog Include File: DW_dp_mult_comb_function.inc

Functional Description

```
z[a_width+b_width-1:0] = DWF_dp_mult_comb (a[a_width-1:0], a_tc,  
                                           b[b_width-1:0], b_tc)
```

```
z (unsigned)    = a (unsigned) * b (unsigned)    if a_tc = 0 and b_tc = 0
```

```
z (signed)      = a (signed)    * b (unsigned)   if a_tc = 1 and b_tc = 0  
                  = a (unsigned) * b (signed)    if a_tc = 0 and b_tc = 1  
                  = a (signed)   * b (signed)    if a_tc = 1 and b_tc = 1
```

For more information about the DesignWare datapath functions, refer to the topic titled [DesignWare Datapath Functions Overview](#).

Related Topics

- [DesignWare Datapath Functions Overview](#)
- [DesignWare Building Block IP User Guide](#)

VHDL Example

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.numeric_std.all;
use DWARE.DW_dp_functions.all;
-- DWARE.DW_dp_functions_arith package if IEEE.std_logic_arith is used

entity DWF_dp_mult_comb_test is
    port (a, b, c      : in  unsigned(7 downto 0);
          a_tc, b_tc  : in  std_logic;
          z            : out unsigned(15 downto 0));
end DWF_dp_mult_comb_test;

architecture rtl of DWF_dp_mult_comb_test is
begin
    z <= DWF_dp_mult_comb (a, a_tc, b, b_tc) + c;
end rtl;
```

Verilog Example

```
module DWF_dp_mult_comb_test (a, b, c, a_tc, b_tc, z);

    input  [7:0] a, b, c;
    input      a_tc, b_tc;
    output [15:0] z;

    // Passes the parameters to the function
    parameter a_width = 8;
    parameter b_width = 8;

    // add "$SYNOPSISYS/dw/sim_ver" to the search path for simulation
    `include "DW_dp_mult_comb_function.inc"

    assign z = DWF_dp_mult_comb (a, a_tc, b, b_tc) + c;

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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